

APEX / SEMS

Sports Event Management System — Comprehensive Architecture

Maharana Pratap Engineering College (MPEC) & Affiliated Campuses • Compliant with AKTU Sports Guidelines

OFFICIAL SYSTEM SPECIFICATION

SANITIZED: ZERO CREDENTIALS

RELEASE v2.4 (2026)

PRIVACY & SECURITY VERIFICATION (ZERO-CREDENTIAL POLICY)

This document is an architectural and functional specification. Under strict institutional security guidelines, all database passwords, master administrative keys, private tokens, payment secrets, and personal identification credentials have been entirely redacted or omitted. System credentials are strictly injected at runtime via encrypted environment variables.

1. EXECUTIVE SUMMARY & PROJECT PURPOSE

The APEX Sports Event Management System (SEMS) is a state-of-the-art, full-stack digital athletics tournament platform engineered specifically for Maharana Pratap Engineering College (MPEC) and all affiliated member institutes under the Maharana Pratap Group of Institutions (MPGI Kanpur). The platform replaces outdated paper dossiers, manual scorekeeping, and unverified spot registrations with a high-integrity, automated, and tamper-proof ecosystem.

- Digital Registration Dossier** Multi-tier athlete and squad registration supporting Individual (Singles), Duo (Doubles/Pairs), and Squad formats with automated player eligibility checks.
- Authoritative Online Payments** Server-side Razorpay order generation with HMAC-SHA256 signature verification, instant PDF pass generation, and Resend transactional email dispatch.
- Court-Side Live Match Portal** Sub-second real-time scoreboards, undo/redo stacks, timeouts, declarations (walkover/disqualified/retired), and telemetry across 12 collegiate disciplines.
- Institutional Tournament Ledger** Deterministic historical fixture scheduling, court timetable management, live standings, and an inter-college championship medal leaderboard.
- Multi-Role Governance Matrix** Strictly isolated portals for Administrators, Super Coordinators, College Heads, Individual Sport Coordinators, and Media/PR Heads.

12

Sanctioned Disciplines

6

Autonomous Portals

100%

Real-Time Telemetry

Zero

Mock Fallback Data

2. FULL-STACK TECHNOLOGY STACK & ARCHITECTURE

The SEMS platform utilizes a modern, resilient, decoupled full-stack architecture built entirely on industry-standard open-source technologies:

Architecture Layer	Technology & Version	Operational Purpose in Codebase
Frontend UI Framework	React 19 (19.2.7)	Concurrent rendering, modern hook lifecycle, strict DOM reconciliation.
Styling & Design Engine	Tailwind CSS v4 + Vite	Veer Legacy design system, hardware-accelerated transitions, dark/light
Client-Side Routing	React Router DOM v7	Declarative routing, navigation guards, role-based portal protection.
API Server Runtime	Node.js LTS (>= 18.0.0)	High-throughput asynchronous I/O with Express.js RESTful routing.
Database & ORM Layer	PostgreSQL & Prisma 7	Type-safe relational modeling, migrations, connection pool via
Security & Rate Limiting	Helmet.js &	HTTP security headers (CSP, frameguard), anti-brute-force IP throttling.
Authentication Engine	JWT & Bcrypt.js	Stateless signed bearer tokens, salted one-way password hashing.
Document & Pass Engine	jsPDF & html2canvas	Vector-sharp collegiate athlete passes, digital receipts, master rosters.
Payment Infrastructure	Razorpay Server SDK	Server-authoritative order creation, HMAC-SHA256 signature verification.
Media & CDN Storage	Cloudinary CDN	High-res tournament photo archive, circular scans, digital certificate
Transactional Messaging	Resend API	Automated participant entry pass delivery with embedded PDF

3. USER ROLES & ACCESS CONTROL MATRIX

Access control is governed by cryptographic JSON Web Tokens (JWT) passed via HTTP Authorization headers (Bearer <token>). Each request to a protected endpoint is intercepted and validated by server middleware against the authenticated user's registered role:

User Role Identifier	Target User Profile	Authorized System Capabilities & Scope
Public Spectator	Unauthenticated public	Public fixtures, live scoreboard, medal standings, circulars,
SPORTS_COORDINATOR	Assigned Sport Specialist	Discipline events, timetable allocation, live court scoring, winner
SUPER_COORDINATOR	Athletic Council Chief	Multi-sport medal point awarding, overall campus champion
COLLEGE_HEAD	Campus Dean / Sports Officer	Strict college-isolated athlete inspection, squad eligibility approvals,
ADMIN	System Director / Registrar	Full institutional control, master data exports (CSV/Excel), audit log
PR_COORDINATOR	Media & Press Council	Championship photo galleries, regulatory press releases, circular

CAMPUS DATA ISOLATION GUARANTEE

College Heads operate under strict relational data partitioning. A College Head authenticated from MPEC can only view, verify, and export rosters representing MPEC athletes. Queries for competing institutions (MIPS, MPCPS, MPCP) are strictly blocked at the database query level.

4. SANCTIONED SPORTS DISCIPLINES & SCORING FORMATS

SEMS embeds custom digital scoring logic and validation constraints for all 12 collegiate sporting disciplines, adhering to official AKTU and national federation standards:

Discipline	Roster Bounds	Official Match Format	Canonical Scorecard Summary
Badminton	1–2 (Singles / Doubles)	3-Set Match (21 pts, 30 cap)	Sets won, individual set breakdowns (e.g. 21-18,
Table Tennis	1–2 (Singles / Doubles)	5-Set Match (11 pts, deuce)	Sets won, detailed point sequence (e.g. 11-8,
Cricket	11–15 Players	T20 Overs, Runs, Wickets	Runs/wickets, completed overs, target chase,
Gully Cricket	5–8 Players	Box Cricket short overs	Total runs, wickets, wall catch rules, run chase
Football	5–11 Players	2 Halves (10-20 min), Extra/Pens	Full-time score, penalty shootout notation (e.g.
Basketball	5–10 Players	4 Quarters, Free throws, Fouls	Cumulative quarter breakdown, final points (e.g.
Volleyball	6–10 Players	3 or 5 Sets (25 pts, final 15)	Sets tally, individual set scores (e.g. 25-21,
Kabaddi	7–12 Players	2 Halves, Raids & Tackles	Raid points, tackle points, all-out counts, total
Kho-Kho	9–12 Players	2 Innings (Chase / Defend)	Points scored per turn, defensive timings
Tug of War	8–10 Players	Best of 3 Pulls	Rounds won, centre marker alignment (e.g. 2-0
Chess	1 Player (Singles)	Classical / Rapid / Blitz	Match points (1-0, 0.5-0.5, 0-1), FIDE result
Athletics	1–4 (Individual / Relay)	Timed Heats & Relays	Official heat timing (mm:ss.ms), podium

5. REGISTRATION, PAYMENT & PASS GENERATION LIFECYCLE

The athlete registration pipeline operates through an authoritative 6-stage verification lifecycle ensuring zero fraudulent entries and instant pass issuance:

- **Stage 1: Category & Format Selection:** The athlete or team captain selects the target sport and format (Singles, Doubles, or Squad). Roster sizes are dynamically constrained based on the sport rules table.
- **Stage 2: Institutional Affiliation & Dossier:** Athletes submit full legal name, father's name, college roll number, date of birth, phone number, and institutional college code.
- **Stage 3: Server-Authoritative Order Creation:** The client requests a payment order via POST /api/public/create-order. The server computes the fee strictly from database event records and creates a Razorpay order, preventing client-side fee tampering.
- **Stage 4: Cryptographic Payment Verification:** Upon transaction completion, the Razorpay payment ID, order ID, and HMAC signature are submitted to POST /api/public/register. The backend validates the signature using crypto SHA256 before registering.
- **Stage 5: Multi-Tier Database Commitment:** Registration details are committed across normalized database models: MasterData, Registration, and individual Athlete records.
- **Stage 6: Digital Pass & Email Dispatch:** A vector-sharp PDF Entry Pass featuring official institutional branding, athlete identity, and unique verification tokens is generated and dispatched via the Resend transactional email API.

6. DATABASE ARCHITECTURE & DATA MODELS

SEMS uses PostgreSQL managed via Prisma 7 ORM. The relational architecture enforces strong relational constraints, cascading deletes, and strict normalization:

Schema Model	Architectural Role & Description in Database
User	System authentication accounts with hashed credentials and assigned UserRole enums.
College	Master institutional directory storing college codes (MPEC, MIPS, MPCPS, MPCP) and official names.

CoordinatorEventItem	Sport event specifications including registration deadlines, entry fees, venues, and coordinator links.
CoordinatorMatch	Scheduled tournament fixtures with court assignments, team linkages, live scores, and match status.
CompletedResult	Archival records of finished matches featuring sport-specific score breakdowns and Player of the Match
MasterData / Registration	Core registration ledger recording payment tokens, registration types (INDIVIDUAL, DUO, TEAM), and
Athlete	Individual athlete records containing full legal names, college roll numbers, contact details, and
AuditLog	Security audit trail logging administrative deletions, master data exports, and manual coordinator

7. SECURITY, PRIVACY & ZERO-CREDENTIAL ARCHITECTURE

SEMS is engineered from the ground up following zero-trust principles and comprehensive data hygiene practices:

- Zero Hardcoded Secrets Policy:** Under no circumstances are production passwords, secret keys, or database URLs committed to source control. All sensitive keys are loaded exclusively at runtime via server environment configurations.
- Bcrypt Cryptographic Password Hashing:** Account credentials stored in the database are hashed using salted Bcrypt rounds, ensuring passwords can never be reversed or exposed even in the event of a database dump.
- Stateless JWT Bearer Token Guarding:** User sessions are verified statelessly via signed JSON Web Tokens. Each request undergoes cryptographic signature and expiration verification.
- Brute-Force & DDoS Mitigation:** The API implements express-rate-limit to throttle excessive requests per IP address, preventing automated credential stuffing, ticket scalping, or server flooding.
- HTTP Security Headers (Helmet.js):** Production HTTP headers enforce strict transport security, Cross-Site Scripting (XSS) filters, clickjacking prevention (X-Frame-Options), and MIME-type sniffing defenses.
- CORS Whitelist Protection:** Dynamic CORS configuration restricts API access strictly to whitelisted frontend hostnames, rejecting unauthorized third-party origins.

8. DESIGN SYSTEM: VEER LEGACY (QUIET STRENGTH)

The APEX visual identity is inspired by institutional dignity, athletic discipline, and geometric clarity. It repudiates garish neon gradients in favor of an authoritative, high-contrast palette:

Design Token Element	Canonical Color / Font	Aesthetic Purpose in Platform
Solid Light Canvas	#FAF9F6 (Ivory)	Primary background in light mode; dignified, clean, paper-like
Atmospheric Dark Canvas	#070A13 (Obsidian)	Primary background in dark mode; deep, focused, low eye-strain
Institutional Violet	#7156A5 / #8B5CF6	Primary action buttons, active navigation indicators, and branding
Championship Gold	#A98B57 / #D2AB45	Reserved exclusively for tournament champions, medals, and
Card Surface Paper	#FFFFFF / #0D101A	High-contrast elevated surfaces for fixtures, statistics, and
Typography: Headings	Cinzel (Display Serif)	Ceremonial typography conveying collegiate heritage and
Typography: Interface	Outfit (Geometric Sans)	Modern, readable interface copy optimized for court-side

9. OPERATIONAL VERIFICATION & MAINTAINER GUIDELINES

For future student maintainers and platform engineers, the repository provides automated build, validation, and database synchronization pipelines:

- **Build Verification:** Run "npm run build" to compile and tree-shake the production bundle. Vite verifies over 2,200 modules in under 4 seconds with zero syntax warnings.
- **Prisma Schema Validation:** Run "npx prisma validate" to check schema integrity and foreign key relationships.
- **Database Synchronization:** Run "npm run db:push" for rapid local iteration, or "npm run db:migrate" to apply versioned migrations.
- **Truthful Empty States:** The platform strictly forbids mock fallback records. If no matches or announcements exist in the database, components render dignified institutional empty states.

OFFICIAL SYSTEM SPECIFICATION • SANITIZED COPY

Directorate of Physical Education & Sports • Maharana Pratap Engineering College (MPGI Kanpur)

All rights reserved © 2026 MPGI Kanpur. Authored for technical handover and institutional compliance.